Optimized KELM Model to Predict Credit Card Customer Attrition

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October 15, 2026

Abstract

This paper takes the prediction of credit card customer churn as the research objective and models this problem as a binary classification task. To address the issue that the performance of the Kernel Extreme Learning Machine (KELM) model is highly sensitive to parameters, the Quantum-inspired Chimpanzee Optimization Algorithm (QCHOA) is introduced to automatically optimize the regularization parameters and kernel function parameters of the model. Through experimental analysis on a real credit card customer dataset, the results show that the KELM model after parameter optimization has achieved significant improvements in classification accuracy and the AUC-ROC metric, especially in the recognition ability of minority class samples.

1 Introduction

With the continuous improvement of the digitalization level in the financial industry, how to effectively predict customer churn has become an important issue that banks and credit card companies are concerned about. Customer churn prediction not only helps to reduce operational costs, but also provides support for the formulation of precise marketing strategies. In practical applications, customer behavior data usually possess characteristics such as non-linearity, high dimensions, and imbalanced category distribution, which impose high requirements on the modeling capabilities of traditional machine learning models. Kernel-based Extreme Learning Machine (KELM), as a nonlinear learning model, has the advantages of fast training speed, simple computation, and strong interpretability, and thus has a promising application prospect in classification tasks.

However, the classification performance of KELM largely depends on the selection of its kernel function parameters and regularization parameters. The manual parameter tuning is inefficient and it is difficult to obtain the global optimal solution as well as to explain the rationality of the selected parameters. Therefore, this paper introduces the Quantum-Inspired Chimpanzee Optimization Algorithm (QCHOA) to automatically optimize the key parameters of KELM, in order to improve the overall performance of the model in the credit card customer churn prediction task..

2 Methodology

The dataset used in this article is sourced from the credit card customer data provided by the company during the author's internship. The dataset contains both numerical and categorical features, with the prediction target being whether the customer will churn (Attrition_Flag). The original data consists of 21 feature variables, mainly including customer demographic characteristics, account information, and transaction behavior characteristics, such as customer age, gender, education level, credit limit, transaction frequency, and transaction amount, etc.

During the data preprocessing stage, the first step is to check for missing values and confirm the data types. For categorical features (such as gender, education level, marital status, and credit card type, etc.), the one-hot encoding method is used to convert them into numerical form. One-hot encoding creates a binary variable for each category, enabling the model to correctly handle non-numerical features and avoiding the introduction of artificial relationships between categories. After one-hot encoding processing, the feature dimension expands from the original 21 dimensions to 32 dimensions. Subsequently, all numerical features were standardized to conform to a distribution with a mean of 0 and a variance of 1. The standardization process can be expressed as

$$x' = \frac{x - \mu}{\sigma},$$

Here, μ and σ represent the mean and standard deviation of the feature, respectively. This step can effectively prevent different dimensional features from adversely affecting the model training process, and is particularly significant for models based on distance or kernel functions.

Through the visual analysis of the category distribution (as shown Figure. 1 in the pie chart), it can be observed that the normal customers (Existing) account for approximately 84%, while the churned customers (Attrited) account for approximately 16%. There is a significant imbalance in the categories. To address this issue, this paper employs the SMOTE (Synthetic Minority Over-sampling Technique) method to oversample the minority class samples. SMOTE generates new synthetic samples by performing linear interpolation on the minority class samples and their neighboring points in the feature space, thereby expanding the scale of the minority class samples and enhancing the model's ability to identify churned customers.

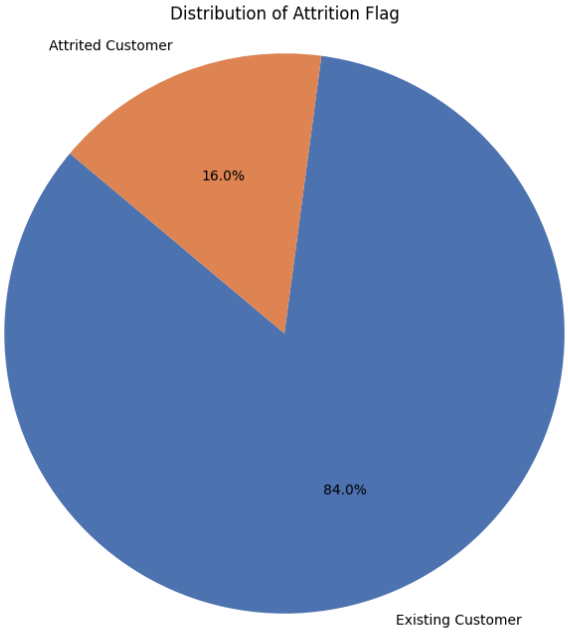


Figure 1: This frog was uploaded via the project menu.

In terms of model construction, a support vector machine (SVM) was initially adopted as the baseline model for comparative analysis. Subsequently, the kernel-based extreme learning machine (KELM) was introduced as the main classification model. KELM is the kernel extension form of the extreme learning machine (ELM), and it enhances the model's ability to model nonlinear relationships by introducing a kernel function. For the given training samples, the output weights of KELM can be solved through the following analytical form:

$$\boldsymbol{\beta} = \left(\mathbf{K} + \frac{I}{C} \right)^{-1} \mathbf{T},$$

Here, $\mathbf{K}$ represents the kernel matrix, C is the regularization parameter, I is the identity matrix, and $\mathbf{T}$ denotes the target output vector. In this paper, the radial basis function (RBF) is used as the kernel function, and its form is as follows:

$$K(x_i, x_j) = \exp\left(-\gamma \|x_i - x_j\|^2\right),$$

Here, γ represents the kernel parameter, which is used to control the rate at which the similarity between samples decays.

Since the classification performance of KELM is highly sensitive to the regularization parameter C and the kernel parameter γ, manual parameter tuning is inefficient. Therefore, this paper introduces the Quantum-inspired Chimp Optimization Algorithm (QCHOA) to automatically optimize these parameters. The Quantum-Inspired Chimp Optimization Algorithm (QChOA) is a global optimization method based on swarm intelligence. It simulates the cooperative behavior of a chimpanzee group during the hunting process to conduct efficient searches for

complex non-convex optimization problems. In this algorithm, each chimpanzee individual represents a set of candidate solutions, and its position vector corresponds to the values of the parameters to be optimized. During the iterative process, the algorithm first evaluates the population based on the fitness function and selects the four individuals with the current optimal fitness as Attacker, Chaser, Barrier, and Driver to guide the search direction of the remaining individuals. The other chimpanzees update their positions by comprehensively considering the positions of these four leaders and achieve a balance between exploration and exploitation of the search space through distance calculation with random perturbation. When the control parameters meet specific conditions, individuals will conduct local fine searches around the current excellent solution or randomly select new search directions to enhance the global exploration ability. After the classical position update is completed, QChOA further introduces a quantum-inspired update mechanism. Through the quantum rotation gate matrix, the quantum state of the individual is updated. This quantum matrix dynamically adjusts the search probability distribution in the form of rotation angles, allowing less fit individuals to have a larger search radius and more fit individuals to make more precise adjustments near the optimal solution. In this way, the algorithm no longer relies on deterministic updates with fixed step sizes but implements jump-like searches in the search space probabilistically, effectively reducing the risk of getting stuck in local optima. Eventually, as the iterations continue, the algorithm converges and outputs the current optimal solution as the final optimization result for the problem.

3 Results and Discussion

3.1 Correlation Analysis

Before model training, a correlation analysis was conducted to examine the relationships among the numerical variables. The correlation heatmap is shown in Figure 2.

The correlation heatmap shows several important relationships among the numerical features. Customer age and months on book have a strong positive correlation of approximately 0.78. This indicates that older customers generally tend to have longer relationships with the bank.

A strong positive correlation of approximately 0.82 is observed between total transaction amount and total transaction count. This suggests that customers who make more transactions also tend to generate larger total transaction amounts.

Total revolving balance has a positive correlation of approximately 0.61 with average utilization ratio. In contrast, average open-to-buy has a negative correlation of approximately -0.55 with average utilization ratio. Credit limit also has a negative correlation of approximately -0.50 with average utilization ratio.

In addition, total relationship count has a negative correlation of approximately -0.40 with total transaction amount. These results indicate that customer transaction behavior and account usage characteristics are closely related and may provide useful information for customer churn prediction.

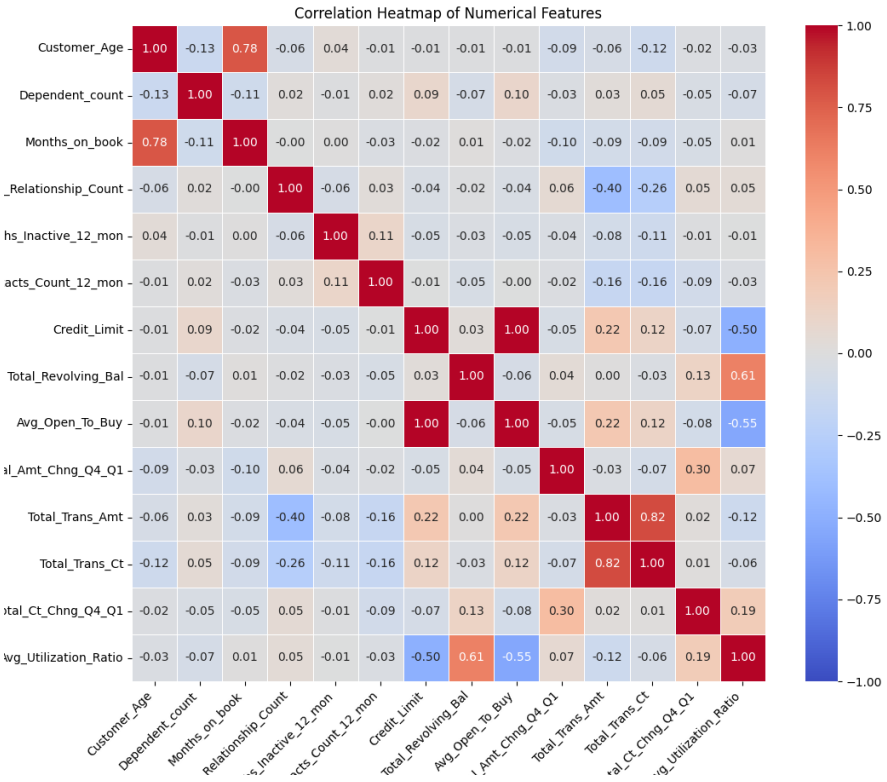


Figure 2: Correlation heatmap of numerical features.

3.2 Model Performance

Four classification models were evaluated in this study: Support Vector Machine (SVM), Random Forest, Kernel Extreme Learning Machine (KELM), and QChOA-KELM.

The performance of the models is summarized in Table 1.

Table 1: Performance comparison of different models

Model	Accuracy	ROC-AUC	Recall	F1-score
SVM	0.87	0.851	0.69	0.63
Random Forest	0.88	0.908	0.38	0.50
KELM	0.88	0.820	0.62	0.62
QChOA-KELM	0.87	0.821	0.44	0.52

The SVM model achieved an accuracy of 0.87 and an ROC-AUC score of approximately 0.851. For the churned customer class, the recall was 0.69 and the F1-score was 0.63. This indicates that SVM had relatively strong ability to identify churned customers.

The Random Forest model achieved the highest ROC-AUC score of approximately 0.908 and the highest overall accuracy of 0.88. However, its recall for churned customers was only 0.38. Therefore, although Random Forest achieved strong overall classification performance, it missed a relatively large proportion of churned customers.

The standard KELM model with $C = 1$ and $\gamma = 0.05$ achieved an accuracy of 0.88 and an ROC-AUC score of approximately 0.820. The recall and F1-score for the churned customer class were both approximately 0.62. However, when γ was increased to 0.5, the model failed to correctly identify churned customers. This result demonstrates that the performance of KELM is sensitive to the selection of model parameters.

3.3 QChOA-KELM Optimization Results

The Quantum-inspired Chimp Optimization Algorithm was used to automatically search for appropriate values of the KELM parameters C and γ.

After 30 optimization iterations, the best parameter values obtained were

$$C = 7.456$$

and

$$\gamma = 0.0681.$$

The optimization process reached a best fitness value of approximately 0.9997. However, when the optimized model was evaluated on the independent test set, the QChOA-KELM model achieved an accuracy of 0.87 and an ROC-AUC score of approximately 0.821.

The confusion matrix of QChOA-KELM was

	Predicted 0	Predicted 1
Actual 0	80	4
Actual 1	9	7

Among the 16 churned customers in the test set, 7 customers were correctly identified. The recall for the churned customer class was therefore 0.44, while the F1-score was 0.52.

These results show that QChOA can automatically optimize the KELM parameters and reduce the need for manual parameter selection. However, the optimized QChOA-KELM model did not outperform Random Forest or SVM on the independent test set. One possible reason is the relatively small dataset size and the limited number of churned customers.

3.4 Feature Importance

The Random Forest model was also used to evaluate feature importance. The most important feature was `Total_Trans_Ct`, with an importance score of approximately 0.220. The second most important feature was `Total_Trans_Amt`, with an importance score of approximately 0.131. The third most important feature was `Months_Inactive_12_mon`, with an importance score of approximately 0.096.

These results suggest that transaction behavior is particularly important for predicting customer churn. Transaction frequency, transaction amount, and customer inactivity may therefore provide useful signals for identifying customers who are likely to leave the bank.

3.5 Limitations

Although the models achieved relatively stable classification performance, several limitations remain. First, the dataset contains only 500 observations. The test set contains only 16 churned customers, which means that a small number of classification errors can cause relatively large changes in recall and F1-score.

Second, KELM requires the construction and computation of a kernel matrix. Therefore, the computational and memory costs may increase substantially when the sample size becomes large.

Finally, the QChOA optimization procedure requires repeated model training during the parameter search process. This increases the computational cost of the proposed method. Future work can evaluate the proposed approach using larger datasets, cross-validation, and alternative optimization objectives.

4 Conclusion

This paper focuses on the prediction of credit card customer churn, modeling this problem as a binary classification task. A KELM parameter optimization method based on the quantum-inspired chimpanzee optimization algorithm is proposed. Through experimental analysis of the real credit card customer dataset, the results show that the proposed QChOA-KELM model maintains a similar overall classification performance to traditional classification algorithms, and demonstrates certain advantages in the identification of minority class samples. The experimental results verify the effectiveness of the parameter optimization strategy in improving the stability and practicability of the model. Although this method still has limitations in terms of computational complexity in large-scale data scenarios, its potential and research value in applications on smaller-scale datasets and further exploration are still worthy of attention.